JAM Gray Paper 口試 100 題

100 questions on the JAM protocol

中英文口述題與短答，每輪五題。

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從 work-package 到鏈上狀態

From work-package to chain state

Q001 §15 §11 §12

從 work-package 開始，誰把工作做成 work-report？報告已獲擔保後，還缺哪些條件才能進入 accumulation？

Who turns a work-package into a work-report, and what must happen after it is guaranteed before it can be accumulated?

> 該 core 的 guarantors 檢查授權、執行各 work-item 的 refine，並簽署報告。Guarantee 納入區塊且通過可用性門檻後，依賴順序與 gas 預算決定何時累積這份報告。
>
> The core's guarantors check authorization, refine the items and sign the report. After inclusion of its guarantee, the report must become available and fit in the affordable prefix of the dependency-ordered candidates.

Q002 §4 §17

Gray Paper 為什麼把 refinement 稱為 in-core computation？當初沒參與計算的驗證者怎麼檢查結果？

Why does the Gray Paper call refinement in-core computation? How can a validator that did not perform the original work check it?

> 每份工作先由指派給該 core 的一小組驗證者執行。稽核者能取回可稽核的 bundle，在相同背景重算 report，檢查 guarantors 擔保的結果。
>
> The validators assigned to a core execute the work first. Auditors can retrieve the auditable bundle, reproduce the report in its specified context and check the guarantors' claim.

Q003 §1 §4 §12

為什麼 Gray Paper 把計算模型稱為 semi-coherent？哪些工作可在不同 core 同時進行，結果又在哪裡匯入同一個鏈上狀態？

Why is JAM called semi-coherent? Which work can run on separate cores, and where are the results integrated into one chain state?

> 不同 work-packages 可以在多個 core 上平行執行 refinement。可累積的報告由各 service 的 accumulate 處理，效果依同一套區塊轉移規則整合到共享狀態。
>
> Separate packages can undergo refinement on different cores in parallel. Services accumulate eligible reports, and their effects are integrated into shared state under one block-transition rule.

Q004 §1 §15 §11

Gray Paper 分開定義驗證者的鏈外工作與鏈上 STF。以 guarantee 為例，產生報告和驗證區塊各要做什麼？

For a guarantee, what belongs to the validator's off-chain work, and what belongs to the block state transition?

> Guarantors 在鏈外執行 refinement 並為 report 簽名。驗塊節點核對 guarantee 的簽名、core 指派、授權與背景等條件，通過後建立待確認的 availability assignment。
>
> Guarantors refine and sign the report off-chain. During block validation, nodes check the guarantee's signatures, assignment, authorization and context before recording a pending availability assignment.

Q005 §2 §17

JAM 沿用 Polkadot 的 ELVES。Backing、approval、inclusion 在 JAM 各對應哪個階段？哪一步把結果帶進鏈上狀態？

Which JAM stages correspond to ELVES backing, approval and inclusion? Which stage applies the results to chain state?

> Backing 對應 guaranteeing，由 guarantors 擔保報告。Approval 對應 auditing，重算並檢查報告。Inclusion 對應 accumulation，將符合條件的工作結果帶入鏈上狀態轉移。
>
> Backing corresponds to guaranteeing a report, approval to auditing its correctness, and inclusion to accumulation. Accumulation applies eligible work results to the on-chain state.

從規格讀出狀態轉移

Reading the state transition

Q006 §11 §4

同一區塊有兩份 guarantee，分別擔保不同的套件，卻指向同一個 core。驗塊節點能留第一份，繼續執行這個區塊嗎？

Two guarantees in one block cover different packages but name the same core. Can a node keep the first and apply the rest of the block?

> 不能。Guarantees 必須按 core index 嚴格遞增，每個 core 最多出現一次。重複就使整個區塊無效。
>
> No. Guarantees must appear in strictly increasing order of core index. A repeated core makes the entire block invalid.

Q007 §4 §10 §11

同一個區塊有 disputes 也有 assurances。檢查 assurance bit 時，要讀哪個版本的 availability assignments？

When a block contains both disputes and assurances, which version of the availability assignments is used to check an assurance bit?

> 讀已套用裁決的中間狀態。若相應報告因 bad 或 wonky verdict 已被清除，該 core 就沒有可供這個 bit 聲明的指派。
>
> Use the post-judgment assignments. If a bad or wonky verdict has already cleared the report, there is no assignment on that core for the assurance bit to attest to.

Q008 §3 §12

把一串待處理的 reports 改存進無序集合，可能在哪裡改變鏈上結果？

How could storing an ordered sequence of reports in an unordered set change the chain result?

> 有 gas 限制時，順序決定哪段前綴能執行。若兩個節點以不同順序取出報告，就可能更新出不同的狀態。
>
> Under a gas limit, order determines which prefix can run. Nodes that iterate over reports in different orders could produce different states.

Q009 §3 §4

兩個開發團隊把區塊驗證步驟排成不同順序，什麼情況下兩者仍可能都正確？

When can two clients evaluate block-transition components in different orders and both be correct?

> 只要都滿足資料依賴，每一步取到規格指定版本的輸入，互不依賴的項目就能交換順序或平行計算。
>
> Independent components can be reordered or parallelized when each receives the exact input versions specified by the dependency graph.

Q010 §4 §5

JAM 的 block extrinsic 分成哪五類資料？各自讓區塊能處理哪件協定工作？

What are the five parts of the block extrinsic, and which protocol task does each serve?

> Tickets 提交出塊票券，disputes 提交裁決與違規證據，preimages 提供已請求的資料。Assurances 聲明驗證者持有對應分片，guarantees 提交 work-reports 及其擔保簽名。
>
> Tickets submit sealing tickets, disputes carry verdicts and offense evidence, and preimages supply requested bytes. Assurances attest to shard possession, and guarantees introduce signed work-reports.

區塊頭能證明什麼

What a block header commits to

Q011 §5

同步節點同時檢查 parent hash 和 prior state root。它們各自固定了什麼？

What do the parent hash and prior state root each establish for a syncing node?

> Parent hash 指定要接在哪個父區塊。Prior state root 承諾執行本區塊的起始狀態，也就是父區塊執行後的狀態。
>
> The parent hash identifies the preceding block. The prior state root commits to this block's starting state, which is its parent's post-state.

Q012 §5 §7

JAM 為什麼把前一個狀態根放進 header？這對出塊流程有什麼好處？

Why does a JAM header carry the prior state root, and how does that help block production?

> 當前 header 可以引用已知的父狀態，把本區塊執行與計算新狀態根安排成管線。新根可由子區塊承諾。
>
> Referencing the parent state lets block computation and Merklization be pipelined. A child header can commit to the newly computed root.

Q013 §5

有人只給你 header 和一份 work-report。要怎麼驗證這份 report 確實被該區塊收錄？

Given a header and a work-report, how can you verify that the block included that report?

> 取得相應的 inclusion proof，依協定的編碼與 Merkle 規則重建承諾，再比對 header 裡的 extrinsic hash。
>
> Obtain the inclusion proof, reconstruct the commitment using the protocol's encoding and Merkle rules, and compare it with the header's extrinsic hash.

Q014 §5 §6

父區塊後跳過好幾個 slot 才出現下一個區塊。節點應怎麼判斷它的時間是否合法？

A child block arrives after several empty slots. How should a node evaluate its timeslot?

> 子區塊的 slot 必須比父區塊晚，且已到本地時鐘允許驗證的時間。中間可以空 slot，因此高度不能代替時槽。
>
> Its slot must be later than its parent's and admissible under the current clock. Slots may be empty, so block height cannot stand in for the slot index.

Q015 §6 §5

同一個 epoch 內，父、子區塊剛好跨過票券提交截止點。若原本的票池還差一張才滿，header 要放 winning-tickets marker 嗎？已集滿時呢？

A parent and child block straddle the ticket deadline in the same epoch. Does the child include a winning-tickets marker if the ticket accumulator is one short? What if it is full?

> 差一張就留空。在這個截止點，只有原本的 ticket accumulator 已滿，才放入依 outside-in 順序排列的中選票券。
>
> If one ticket is missing, the marker is empty. At that crossing, a full prior accumulator produces the winning tickets in outside-in order.

誰有資格出塊

Who gets to author blocks

Q016 §6 §19

Safrole 已經指定某個 slot 的作者。為什麼之後還要 GRANDPA？

If Safrole assigns an author to a slot, what work remains for GRANDPA?

> Safrole 管理鏈如何延長，仍可能出現競爭分支。GRANDPA 用驗證者投票確定一段共同歷史。
>
> Safrole governs chain extension, where competing branches can still arise. GRANDPA uses validator votes to finalize a common history.

Q017 §6 App. G

為什麼可以先公布中選票券，又不必提前公開每張票屬於哪個驗證者？

How can winning tickets be public while the identities of their holders remain hidden in advance?

> Ring VRF 讓驗證者證明自己是合格集合的一員並產生可驗證的票券。持票者到指定 slot 才用相應證明封塊，較難被預先鎖定。
>
> A ring VRF proves eligible membership and produces a verifiable ticket without revealing which member owns it. The holder later proves entitlement when sealing the block.

Q018 §6

換 epoch 時，如果票券競賽未能產生可用的完整排程，鏈靠什麼繼續出塊？

How can the chain continue if the ticket contest fails to yield a usable schedule for the next epoch?

> Safrole 依規則從驗證者金鑰與既有熵產生 fallback 排程。各節點可獨立算出同一組有資格封塊的金鑰。
>
> Safrole derives a fallback schedule from validator keys and recorded entropy. Nodes independently recover the same eligible sealing keys.

Q019 §6 §12

跨到新 epoch 的區塊，active validator set 從哪裡來？同一塊 accumulation 才寫入的 staging keys 何時才有作用？

At an epoch boundary, where does the active validator set come from, and when can staging keys written by that block's accumulation take effect?

> 先前的 pending set 成為 active，這次輪替用區塊前的 staging keys 準備新的 pending set，並過濾違規者。該塊 accumulation 更新的 staging keys 留給後續輪替。
>
> The previous pending set becomes active. Rotation prepares the new pending set from prior staging keys with the offender filter. Staging keys written during this block's accumulation are available for later rotations.

Q020 §6 §11

分配驗證者任務需要隨機性。不同節點怎麼拿到相同的隨機來源？

How do nodes obtain a common source of randomness for validator assignments?

> 區塊的 VRF 輸出按規則混入 entropy accumulator。協定保留歷史 epoch 的熵，指定任務使用其中哪一份。
>
> Block VRF outputs feed a shared entropy accumulator. The protocol retains epoch snapshots and specifies which snapshot seeds each assignment.

報告怎麼連回歷史

How reports refer to history

Q021 §7 §11

一份新 report 聲稱它參照某個近期區塊。節點用什麼來核對這個 anchor？

How does a node check a new report's claim about its recent anchor block?

> 比對近期歷史中保存的 header hash、slot、狀態根與累積輸出承諾。這讓引用過期或捏造近期背景的報告無法通過該檢查。
>
> The node matches the header hash, slot, state root and accumulation-output commitment against recent history. A stale or fabricated recent context fails this check.

Q022 §7 §11 §14

後一份工作聲稱「套件 P 產生的 segments root 是 S」。節點為什麼能核對這組對應？

How can a node verify a claim that package P produced segments root S?

> 近期歷史保留已報告的 package hash 到 segments root 的對應，當前區塊的報告也可提供依據。節點用這些紀錄核對引用。
>
> Recent history records package hashes and their reported segment roots. Reports in the current block can also establish the mapping used for this check.

Q023 §11 §9 §7

一個 work-report 同時有 anchor 和 lookup anchor。哪個固定近期鏈背景？重跑 refinement 查 preimage 時又要看哪個？

What is the difference between the anchor and the lookup anchor in a work-report's refinement context?

> Anchor 讓節點以近期歷史核對報告引用的區塊與承諾。Lookup anchor 固定查詢服務程式與 preimages 的歷史時點，讓重跑能看到當時的資料。
>
> The anchor ties the report to a recent block and its commitments. The lookup anchor fixes the historical view for code and preimage lookups during reproduction.

Q024 §7 §12

在 recent history 找到套件 P 的 hash，就能斷定它已被累積嗎？

Does finding a package hash in recent history prove that the package was accumulated?

> 它能證明 P 曾被報告。要確認已經累積，還要查 accumulation 的完成紀錄，兩者追蹤的階段不同。
>
> It establishes that P was reported. Accumulated-package history records which packages actually reached accumulation.

Q025 §7 §12 §18

外部系統只想驗證某個服務發布的累積輸出。JAM 如何把這類輸出留成可供證明的歷史？

How does JAM preserve a provable history of outputs published by services during accumulation?

> 每個區塊把服務識別碼與輸出雜湊組成序列，對它取根，再把根追加到 accumulation output log。之後可針對那筆輸出建立包含證明。
>
> Each block commits to its sequence of service IDs and output hashes. The block-level root is appended to the accumulation output log, supporting later inclusion proofs.

誰可以使用 coretime

Who may use coretime

Q026 §8 §4

Authorization 把 coretime 的分配和哪個決定分開？對一份具體 work-package，誰判斷它能否在指定 core 執行？

Which decision does authorization separate from allocating coretime, and who decides whether a particular package may run?

> Coretime 可以先分配。具體工作送達時，該 core 接受的 authorizer 依 work-package 及其 token 執行授權判斷。
>
> Coretime can be allocated first. When a particular package is submitted, an authorizer accepted for the core evaluates the package and its token.

Q027 §8 §15

一份工作聲稱自己有權使用某個 core。誰提供授權證據，誰判斷，成功後會留下什麼？

Who supplies evidence that a package may use a core, who evaluates it, and what does success produce?

> 套件帶入 token，authorizer 程式執行授權判斷。成功時產生 trace，供後續報告描述這次授權。
>
> The package supplies a token. The authorizer evaluates the request and, on success, returns a trace that is carried in the report.

Q028 §8 §11

執行 authorizer 得到成功結果後，為什麼鏈上還要檢查該 core 的 authorizer pool？

Why check the core's authorizer pool on-chain after the authorizer has approved the package?

> 程式成功表示工作符合該 authorizer 的規則。鏈上還要確認這個 authorizer 的 hash 當時確實獲准代表該 core 授權。
>
> Execution shows that the package satisfied that authorizer's rules. Pool membership shows that this core actually accepted that authorizer at the relevant time.

Q029 §8 App. I

Authorizer pool 已滿八項，現在要從 queue 補入一項。有使用該 core 的 guarantee 和沒有 guarantee 時，各移除哪一項？

A core's authorizer pool is full. When a queue entry is appended, which entry leaves if the block has a guarantee for that core, and which leaves if it does not?

> 有 guarantee 時，先移除其 authorizer hash 在 pool 中最早的匹配項，再補入新項。沒有 guarantee 時直接補入，超過八項就淘汰最前面的舊項。
>
> A guarantee first consumes the oldest pool entry matching its authorizer hash, making room for the new entry. Without a guarantee, appending overflows the full pool and evicts its oldest entry.

Q030 §8 §9 §12

某個 core 想改未來接受工作的政策。誰能改排程，這項變動何時進入 pool 計算？

Who can change a core's future authorization schedule, and when does the pool use it?

> 該 core 的特權 assigner 可在 accumulation 改 authorizer queue。區塊計算新 pool 時，按目前 slot 從更新後的 queue 取值。
>
> The core's privileged assigner may change the queue during accumulation. The pool update selects the current slot's entry from that updated queue.

服務帳戶、儲存與特權

Service accounts, storage and privileges

Q031 §9 §4

給定 service ID，服務帳戶記錄哪些狀態？節點如何從帳戶裡的 code hash 找到可執行的程式？

What does a service account record, and how does a node obtain its executable code from the account's code hash?

> 用 service ID 查到的帳戶包含 storage、balance、code hash 與 preimage 資料等。節點用 code hash 查對應的 preimage，再解出 metadata 與程式碼。
>
> The service ID identifies an account with storage, balance, a code hash and preimage data. Looking up the code hash gives the encoded metadata and code, if the required preimage is present.

Q032 §9

Service account 的 threshold balance 如何反映儲存占用？為什麼只知道儲存的位元組數還不夠？

How does a service account's threshold balance reflect its footprint? Why is the byte count alone insufficient?

> 規格把基本押金、項目數成本與位元組成本加總，再扣除 storage credit，結果最低為零。因此項目數本身也會影響門檻。
>
> The threshold combines a base deposit with per-item and per-byte charges, subtracts storage credit, and floors the result at zero. The number of items affects the threshold independently of their size.

Q033 §9 §14

服務剛升級程式。要重跑一份升級前建立的 work-item，怎麼知道該執行哪份程式碼？

A service has upgraded. How can a validator identify the code for a work-item created before the upgrade?

> Work-item 指定 service ID 和 code hash。驗證者依 lookup anchor 的歷史視角取得該 hash 的程式原文。
>
> The work-item names the service and code hash. The validator looks up the matching code preimage using the historical view fixed by the lookup anchor.

Q034 §12 §9

Preimage 在 prior state 已被請求，但服務在本區塊的 accumulation 撤銷了請求。它能通過哪一步檢查？最後會寫入嗎？

A preimage was requested in the prior state, but the service drops the request during accumulation. Which check can pass, and are the bytes stored?

> 若 hash、長度吻合且原請求待供應，它通過以 prior state 作的請求檢查。最後整合時再查當下帳戶，已撤銷的請求會被略過。
>
> Matching bytes can pass the prior-state solicitation check if the request was pending. Final integration checks the current account again and skips the provision if the request has been dropped.

Q035 §9 §12

哪些服務可以改 staging validator keys，哪些可以改 authorizer queue？權限是全體 core 共用，還是按 core 指派？

Who may change the staging validator keys and the authorizer queues? Which authority is assigned per core?

> Delegator 可更新整組 staging validator keys。每個 core 則有自己的 assigner，負責該 core 的 authorizer queue，兩種角色都記在 privileges 狀態中。
>
> The delegator can update the staging validator keys. Each core has an assigner for its own authorizer queue. Both roles are identified in the privileges state.

出現爭議時，鏈採信什麼

How a dispute becomes chain evidence

Q036 §10 §17

一名稽核者認為 report 算錯了。它簽署的 judgment 和可以納入區塊的 verdict 有何差別？

How does one auditor's signed judgment differ from a verdict that a block can include?

> Judgment 是單一驗證者對報告的判斷。Verdict 彙整符合規則的驗證者簽名及票數，讓所有節點可重算裁決。
>
> A judgment is one validator's signed assessment. A verdict collects the required judgments so all nodes can verify the signers, votes and outcome.

Q037 §10

同一份爭議報告隔天又被提交。保留 good、bad、wonky 這些歷史紀錄有什麼作用？

Why retain good, bad and wonky report histories when a disputed report might reappear later?

> 節點可以辨認過去已判定有效、無效或無法確定的報告，依既有裁決拒絕重複爭議或不合格的再提交。
>
> Nodes can recognize a report already judged valid, invalid or indeterminate. The recorded outcome supports rejecting repeat verdicts and inadmissible resubmissions.

Q038 §10 §11

一份 report 還在某個 core 上等 assurance，此時鏈上出現 bad 或 wonky verdict。排程要怎麼處理？

A report is still awaiting assurance when a bad or wonky verdict is recorded. What happens to the assignment?

> 該 core 上與報告相符的 pending availability assignment 會被清除，這份報告不會循這個指派進入可用報告序列。
>
> The matching pending availability assignment is cleared. That assignment can no longer move the report into the sequence of newly available reports.

Q039 §10

一個人簽了錯誤的 guarantee，另一個人簽了與裁決相反的 judgment。鏈上如何區分責任？

How does the chain distinguish guaranteeing an invalid report from issuing a judgment that contradicts the verdict?

> Culprit 證據指出誰擔保了無效報告。Fault 證據指出誰簽了與有效性裁決相反的 judgment。兩者都要能驗簽並對應到報告。
>
> Culprit evidence identifies a guarantor of an invalid report. Fault evidence identifies a signer of a contradicting judgment. Both require signatures tied to the report.

Q040 §10 §13

已查到違規驗證者的金鑰。從認定違規到實際處分，JAM 和上層制度各負責什麼？

After an offending validator is identified, what does JAM record and who can impose economic penalties?

> JAM 保存可驗證的裁決與違規金鑰，讓節點對事件有共同紀錄。上層的驗證者選擇或 staking 系統可據此執行懲罰。
>
> JAM records verifiable verdicts and offending keys. A higher-level validator-selection or staking system can act on that evidence to impose penalties.

報告上鏈與資料可用

Reporting and data availability

Q041 §11 §15

一份 report 有足夠的 guarantor 簽名。區塊作者為什麼還要看它對應的 core 是否空著？

Why must a block author check that a core is free even when a report has enough guarantor signatures?

> 每個 core 同時只保留一個待確認的 availability assignment。前一份指派尚未清除時，新的 guarantee 不能占用它。
>
> Each core has room for one outstanding availability assignment. A new guarantee can take the core only after the previous assignment has cleared.

Q042 §11 §16

「已有人擔保這份 report」和「它已 available」分別需要什麼證據？

What evidence distinguishes a guaranteed report from an available report?

> Guarantee 由該 core 指派的 guarantors 簽署。成為 available 則需在區塊的 assurances 中，超過三分之二的 active validators 標記該 core 的資料已取得。
>
> A guarantee carries signatures from assigned guarantors. Availability requires assurances marking that core from more than two-thirds of active validators in the block.

Q043 §11

可用性聲明只列出哪些 core 的資料已收到。為何簽名還要綁定 parent hash？

Why bind an assurance to the parent hash when its bitfield already names the cores?

> 同一個 core 在不同鏈背景可能掛著不同的報告。簽入 parent hash 能讓節點把聲明對準正確的指派，並拒絕跨背景重放。
>
> A core can refer to different reports in different chain contexts. Signing the parent hash ties the bitfield to the intended assignments and prevents replay across those contexts.

Q044 §11 §15

Guarantors 簽完後剛好輪替，報告送到區塊作者時該用哪一組指派來驗簽？

Guarantors rotate just after signing. Which assignment should a block author use to verify their report?

> 依 guarantee 的 timeslot 找出當時該 core 的指派，再套用允許的輪替窗口。符合條件的上一輪簽名也能被核對。
>
> Use the assignment for the guarantee's timeslot and enforce the admitted rotation window. Signatures from the immediately preceding rotation may still qualify.

Q045 §11

某個 core 的報告一直收不到足夠的 assurances。協定怎麼讓 core 能繼續接工作？

How can a core return to service if its pending report never receives enough assurances?

> Pending assignment 到達 timeout 會被清空，core 可以再接新 guarantee。過期這件事本身不會讓舊報告變成 available。
>
> The pending assignment is cleared at timeout, freeing the core for a new guarantee. Expiry alone does not establish availability for the old report.

報告何時能進入 accumulation

When reports can be accumulated

Q046 §12 §14

Report B 依賴套件 A。A 已在先前區塊累積，和 A 在本區塊才排到 B 前面，排序時怎麼區分？

Report B depends on package A. How does ordering handle A having been accumulated earlier versus A being selected ahead of B in this block?

> 先用 accumulated history 移除過去已滿足的依賴。排本區塊的候選順序時，先選出的 A 也能解鎖 B。區塊完成後，只有實際執行的前綴進入歷史。
>
> Accumulated history clears dependencies satisfied in earlier blocks. Selecting A earlier in this block's candidate order can also unlock B. Only the prefix actually processed is recorded in history.

Q047 §12

依賴排序已把 A 和 B 都列進候選，但本區塊只處理了 A。哪一個能記進完成歷史？

Dependency ordering selects both A and B as candidates, but the block processes only A. Which one enters accumulated history?

> 只有 A。候選順序讓節點知道可依什麼次序執行，寫入完成歷史的是實際處理的 report 前綴所對應的套件。
>
> Only A. Candidate order establishes an execution order. History records the packages in the prefix of reports actually processed.

Q048 §12

本輪剩餘 gas 放不下候選序列的第一份 report，卻放得下第二份。外層 accumulation 怎麼選？

This round can afford the second candidate report but not the first. Which report prefix does outer accumulation select?

> 它取能放進預算的最長連續前綴，所以這一輪選到的 report 數是零。後續輪次若還有工作，仍從同一個序列頭繼續考慮。
>
> It takes the longest affordable contiguous prefix. Here that prefix contains zero reports. If other work leads to another round, the remaining sequence still begins with the first report.

Q049 §12

第 12 章的外層、平行層和單一服務層，各負責什麼？哪一層決定本輪取哪些 reports？

What do the outer, parallel and single-service accumulation functions each do? Which one chooses the reports for a round?

> Δ+ 按 gas 選前綴並安排後續輪次。Δ* 在同輪執行各 service 並合併結果。Δ1 為單一 service 準備 gas 和輸入，呼叫 accumulate。
>
> The outer function selects the affordable prefix and schedules later rounds. The parallel function merges services' results from the same round input state. The single-service function prepares gas and inputs and invokes accumulate.

Q050 §12

這個區塊沒有任何 work-item 指向服務 S。它還可能因哪些事情執行 accumulate？

What can trigger accumulation for service S when no work-item targets it in this block?

> 它可能收到 deferred transfer，或本區塊起始的 always-accumulate 配置給它基本 gas。後者在第一輪帶入。
>
> S can receive a deferred transfer or be listed for a base gas allowance under always-accumulate. That allowance is supplied in the block's first round.

同一區塊裡的輪次與紀錄

Rounds and accounting within a block

Q051 §12

服務 A 在一輪 accumulate 中對 B 發出 deferred transfer。誰把它交給 B，B 最早在哪一輪收到？

Service A emits a deferred transfer to B during accumulation. How is it delivered, and when can B first receive it?

> 本輪先合併各服務產生的 transfers。外層再把它們當作同一個區塊下一輪的輸入，B 的呼叫便能收到這筆 transfer 與其 gas allowance。
>
> Transfers are collected when the current round is merged. The outer function passes them to the next round in the same block, where B can receive the transfer and its gas allowance.

Q052 §9 §12

Manager 在本輪把某個 core 的 assigner 從 A 換成 B。本輪採用誰輸出的 queue，B 何時能行使新角色？

The manager replaces core assigner A with B in this round. Whose queue is merged now, and when can B exercise the role?

> 本輪用起始時持有角色的 A 所輸出的 queue。角色變更合併進新狀態後，後續輪次才會把 B 當作該 core 的 assigner。
>
> This round takes the queue from A, the holder at its start. Following rounds see B as the assigner after the role update is merged.

Q053 §13

想比較驗證者一段時間的貢獻，又想分析某個區塊的服務負載，為何要看不同時間尺度的統計？

Why use different time windows to compare validator contributions and diagnose one block's service load?

> Validator activity 按 epoch 累積，保留當期與前一期紀錄。Core 和 service 的統計則描述單一區塊。
>
> Validator activity accumulates over an epoch, with current and previous records retained. Core and service statistics describe individual blocks.

Q054 §10 §13

如果 JAM 沒有直接替每次工作發獎勵，為什麼鏈上仍要記錄驗證者活動？

Why record validator activity on-chain if JAM does not pay each reward directly?

> 出塊、擔保與 assurance 等可觀察活動，能形成各節點一致核對的紀錄。上層 staking 系統可利用這些紀錄決定獎勵。
>
> Block production, guarantees and assurances leave objective evidence in the chain. A higher-level staking system can use those records when allocating rewards.

Q055 §12 §13

某服務在這個區塊有 refinement gas 紀錄，accumulation 的工作量卻是零。怎麼解釋？

How can a service show refinement gas in a block but no accumulation work?

> Refinement 統計來自本區塊新報告的工作。Accumulation 統計看這個區塊實際處理的項目與轉帳，同一份工作的兩個階段可能落在不同區塊。
>
> Newly reported work contributes refinement statistics. Accumulation statistics follow the items and transfers actually processed, which can occur in a later block.

Work-package 裡究竟裝什麼

What a work-package carries

Q056 §14 §11

一個 work-item 如何指定要執行的服務？Payload 和 manifest 各為 refinement 提供哪些資訊？

How does a work-item identify the service to run, and what do its payload and manifest supply to refinement?

> Work-item 指定一個 service ID 與 code hash。Payload 交由該服務解讀，manifest 則描述匯入的 segments、外部資料的雜湊與長度，以及預計匯出的數量。
>
> A work-item names one service ID and code hash. The service interprets its payload. Its manifest identifies imported segments, extrinsic data hashes and lengths, and the declared export count.

Q057 §11 §12 §14

一個被成功評估的 work-package 裡有三個 items，分別給兩個服務。會產生幾份 report，結果怎麼分送？

A successfully evaluated package contains three items targeting two services. How many reports result, and how are the outputs routed?

> 對這次評估產生一份 work-report，內含每個 item 各自的 digest。累積時，這些結果依目標 service 分組送入執行。
>
> That evaluation yields one report containing a digest per item. Accumulation groups the relevant results by their target service.

Q058 §11 §14

某個 item 的 refine 耗盡 gas。這會怎麼反映在報告裡，審核報告正確性時該比對什麼？

An item runs out of gas during refinement. How does that affect its digest and the check on report correctness?

> 該 digest 可以記錄 out-of-gas 這類工作錯誤。審核時比對重跑的結果是否與報告一致，正確記錄的執行錯誤也是可驗證的結果。
>
> The digest can record an out-of-gas work error. An auditor checks that reproduction yields the reported outcome, including a correctly reported execution failure.

Q059 §12 §14

為什麼同一個 work-item 要分別聲明 refinement 和 accumulation 的 gas limit？

Why give one work-item separate gas limits for refinement and accumulation?

> 兩個階段使用不同執行環境。前者限制 core 上的計算，後者讓節點安排之後更新共享狀態所需的預算。
>
> The stages run in different execution contexts. One limit bounds in-core computation, while the other budgets the later on-chain state update.

Q060 §14 §16

服務 B 只需要服務 A 先前產出的一段大型資料，下一份工作怎麼取得並驗證它？

How can service B reuse and verify one piece of a large output previously produced by service A?

> A 的工作可匯出 segments，由 D^3L 分散保存，work-report 承諾其 segments root。B 的工作引用片段，取回後用證明對照該 root。
>
> A's work exports segments to D^3L and commits to their root in its report. B's work references the segment, retrieves it and checks its inclusion proof.

Guarantor 實際承諾什麼

What a guarantor commits to

Q061 §14 §15

Guarantor 的簽名交給區塊作者後，其他人能從這個簽名追究哪一項承諾？

What claim can others hold a guarantor accountable for after its signature reaches a block author?

> 簽名綁定具體 work-report 的雜湊，表示這名驗證者擔保該報告忠實反映規定的計算。日後可重算並核對這個承諾。
>
> The signature binds the guarantor to a particular report hash and its claimed computation. Others can reproduce the work and hold the signer to that claim.

Q062 §11 §15

某個 active core 的一名 guarantor 暫時離線，另外兩位算出相同的報告。這份工作還有機會上鏈嗎？

One guarantor for an active core is offline. Can the other two still produce an includable guarantee if they agree?

> 每個 active core 指派三位驗證者，兩位指派成員的有效簽名就可滿足簽名門檻。報告仍須通過其他納入條件。
>
> Three validators are assigned per active core. Two valid assigned signatures can meet the signing threshold, provided the other inclusion rules also hold.

Q063 §8 §11 §15

Guarantor 準備花很多算力跑 refine。有哪些便宜的檢查值得先做，為什麼？

Which cheap checks should a guarantor make before an expensive refinement, and why?

> 先檢查授權、對應 core 的 pool 與報告背景是否可能符合納入規則。即使計算結果正確，不合納入條件也可能白做。
>
> Check authorization, core-pool membership and contextual eligibility first. An accurately computed report can still be ineligible for inclusion.

Q064 §11 §15

同一個 core 的 guarantors 各自拿到不同套件。如果互不協調，可能浪費什麼工作？

What work is wasted if guarantors on the same core independently choose different packages?

> 同一個 guarantee 需要對同一份 report 的簽名。提早分享候選套件能讓同 core 的成員評估同一份工作，再彙集相符的簽名。
>
> A guarantee needs signatures on the same report. Sharing a candidate package early helps guarantors evaluate the same work and combine matching attestations.

Q065 §14 §15 §16

Guarantor 已經把 report 送給區塊作者。為什麼它還要向驗證者網路分發資料？

After giving the report to a block author, why must guarantors also distribute data to the validator network?

> 其他驗證者需要資料來稽核，後續工作也可能要匯入輸出。Guarantors 因此把可稽核的 bundle 與匯出 segments 做糾刪編碼並分發分片。
>
> Auditors need the inputs, and later work may need the exports. Guarantors erasure-code and distribute both the auditable bundle and the exported segments.

資料能不能取回

Can the data be recovered?

Q066 §16 App. H

驗證者準備對一個 core 簽 assurance。它在簽名前要親自取得並核對什麼？

What must a validator obtain and check before signing an assurance for a core?

> 它要持有分配給自己的 bundle 分片及相應的匯出片段分片，並核對對應位置的證明，才能聲明持有資料。
>
> It must hold its assigned bundle shard and the corresponding shards of exported segments, and verify the proofs for their positions before claiming possession.

Q067 §14 §16

對同一份 work-package，稽核用的 bundle 與留給後續工作的匯出資料，為什麼有不同保存需求？

Why do the auditable bundle and a package's exported data have different retention needs?

> Bundle 服務近期的重跑稽核。匯出資料屬於較長期的 D^3L，其他工作日後仍可匯入，規格要求把相關分片保留至少 28 天。
>
> The bundle serves near-term auditing. Exports go into the longer-lived D^3L so later work can import them. Their shards are retained for at least 28 days.

Q068 §14 §16

舊工作匯出的資料還在保存期內，但驗證者已經換屆。要向哪一批人請求分片？

An export is still within its retention window, but validators have rotated. Which set should a node fetch shards from?

> 依當時對該資料作 assurance 的驗證者集合與分片位置取得資料。現在的 active set 可能已經換人，甚至規模也可能不同。
>
> Use the validator set that assured the data and its shard indices. Membership and set size may both have changed since then.

Q069 §14 §16 App. H

一個有 Merkle 證明的分片，和足夠多個能還原資料的分片，分別解決哪個問題？

What does proving one shard establish, and what does collecting enough distinct shards establish?

> 證明用來驗證分片確實位於承諾的位置。收集達門檻的不同有效分片，糾刪解碼才可還原整份資料。
>
> The proof authenticates the shard at its committed position. A sufficient set of distinct valid shards lets the erasure decoder recover the full data.

Q070 §11 §16 App. H

一份報告需要超過三分之二的 availability assurances，但重建時只取部分分片。這兩個門檻各保護什麼？

Why can availability require a validator supermajority even though recovery uses only a subset of shards?

> 聲明門檻是協定認定資料已分散持有的條件。重建門檻是編碼的數學性質，決定需要多少個不同的有效分片才能解碼。
>
> The assurance threshold is the protocol's evidence of distribution. The recovery threshold is a property of the code that determines how many distinct valid shards suffice to decode.

稽核職責與 audited 判定

Audit duties and the audited condition

Q071 §17 App. G

Auditing 是驗證者自行決定要不要做嗎？它如何證明自己被選中，發出 announcement 後又承諾了什麼？

How are auditing duties determined, and what does a validator commit to when it announces an audit?

> 驗證者依 VRF 規則決定自己應檢查哪些 reports，並隨公告提供選取證明。已公告的 audit 必須完成，且須發布簽署的 judgment。
>
> Each validator derives its audit duties using the protocol's VRF rules and proves its selection when announcing. An announced audit must be completed and a signed judgment published.

Q072 §17 §20 App. G

少數 guarantors 先算，之後每份 report 平均也只被抽查約三十次。為何這能提高對正確性的信心？

Why can roughly thirty sampled audits give confidence in a report initially computed by only a few guarantors?

> 可驗證的隨機選取讓攻擊者難以預先鎖定稽核者。在協定安全假設下，抽中的人全是同夥的機率很低，誠實稽核者的負評會觸發更廣泛的檢查。
>
> Verifiable random selection makes assignments hard to predict in advance. Under the security assumptions, an entirely malicious sample is unlikely, and an honest negative judgment triggers wider checking.

Q073 §17 §20

全規模設定下，每位驗證者初始抽十個 core。為什麼一份 report 平均約有三十次稽核？

At full scale, each validator initially samples ten cores. Why does a report receive about thirty audits on average?

> 兩個數字的分母不同。按 1,023 位驗證者、341 個 active cores 的模型，平均每個 core 被選中的次數是 1,023 × 10 ÷ 341，也就是 30。
>
> One figure is per validator and the other per report. In the 1,023-validator, 341-active-core model, 1,023 times 10 divided by 341 gives 30 expected selections per core.

Q074 §10 §17

已宣布的稽核遲遲沒有回覆，和有人明確發出負面 judgment，接下來的處理有何不同？

How does a missing announced judgment differ from an explicit negative judgment in subsequent auditing?

> 缺席會讓後續 tranche 依 VRF 規則選出額外稽核工作。看到負面 judgment 時，所有驗證者都應就該報告稽核並提出判斷。
>
> A no-show can assign additional checks through VRF rules in later tranches. A negative judgment requires all validators to audit that report and issue a judgment.

Q075 §17 §19

節點何時可把一份 work-report 視為 audited？要把整個 block 視為 audited，還需要哪些 reports 也通過？

When can a node consider a work-report audited, and which reports must pass for the block to be audited?

> 一種情況是沒有負評，且某個 tranche 的應稽核者都已給正評。另一種是全體驗證者中超過三分之二給正評。本區塊新變為 available 的 reports 都通過，區塊才算 audited。
>
> A report qualifies with no negative judgments and all required positive judgments for some tranche, or with positive judgments from over two-thirds of validators. Every newly available report in the block must qualify.

最終性與跨鏈證明

Finality and external proofs

Q076 §18 §19

GRANDPA 已經確定區塊，為什麼還有 BEEFY 簽章？驗證者在什麼時候簽、簽的是哪個承諾？

After GRANDPA finality, when do validators produce BEEFY signatures, what do they sign, and why?

> 驗證者匯入 finalized block 後，以 BLS 簽署相應的 BEEFY 輸出歷史承諾。簽名可聚合，第三方系統因此能驗證精簡的最終性證據。
>
> After importing a finalized block, validators sign the corresponding BEEFY output-history commitment with BLS keys. Aggregation gives third-party systems compact evidence of finality.

Q077 §7 §18 App. E

從一筆服務輸出到 BEEFY 簽署的歷史承諾，中間有哪兩層包含關係？

What two inclusion relationships connect a single service output to the history commitment signed by BEEFY?

> 先證明服務 ID 與輸出 hash 包含在該區塊的輸出根。再證明這個區塊輸出根包含在已簽署承諾所涵蓋的 accumulation output log。
>
> First prove that the service ID and output hash belong to the block's output root. Then prove that the root belongs to the accumulation output log covered by the signed commitment.

Q078 §6 §17 §19

節點看到兩個競爭鏈頭。比較哪條鏈比較好之前，要先排除哪些候選？

Before comparing two chain heads, which candidates must a node rule out?

> 候選須延續已 finalized 的區塊，通過本地的 audited 條件，且未確定的路徑不可包含已觀察到的同 slot 雙重出塊。
>
> A candidate must descend from the finalized block, satisfy the local audit condition and have no observed equivocation in its unfinalized ancestry.

Q079 §6 §19

兩條分支都符合候選條件，較長的一條卻有許多 fallback blocks。最佳鏈怎麼選？

Both forks are acceptable, but the longer one has many fallback blocks. How is the best chain chosen?

> 比較候選歷史中以一般 slot-sealer ticket 封成的區塊數，選使該數量最大的鏈頭。區塊總數本身不足以決定結果。
>
> Among acceptable heads, maximize the number of ticket-sealed blocks in the ancestry. A longer fork with more fallback blocks need not win.

Q080 §5 §19

兩個驗證者投給相同的 header，算出的 post-state root 卻不同。GRANDPA 會把票算在一起嗎？

Two validators vote for the same header but attach different post-state roots. Do those votes support the same target?

> JAM 的 GRANDPA 投票同時承諾 header 和執行後的狀態根。根不同就屬於不同投票目標，這會暴露執行結果的歧異。
>
> A JAM GRANDPA vote binds the header together with its post-state root. Different roots make different voting targets and expose disagreement about execution.

從規則看 JAM 的分工

How JAM divides responsibility

Q081 §20 §5

Gray Paper 如何區分 spatial parallelism 與 temporal parallelism？兩者怎麼讓 JAM 在同一段時間完成更多計算？

How do spatial and temporal parallelism each contribute to JAM's performance model?

> 空間平行讓不同 core 同時處理獨立工作。時間平行則讓不同區塊的計算階段重疊，例如 header 引用 prior state root，使區塊執行與 Merklization 可以管線化。
>
> Spatial parallelism puts independent work on different cores at once. Temporal parallelism overlaps stages across blocks. A header's prior state root, for example, enables block computation and Merklization to be pipelined.

Q082 §14 §12

套件 B 匯入 A 的 segment，沒有另列 prerequisite。用 A 的 package hash 引用，或直接給 segments root，哪種寫法會讓 accumulation 把 A 當成 B 的依賴？

B imports a segment from A without listing A as a prerequisite. Does importing by A's package hash make B depend on A for accumulation? What if B uses the segments root directly?

> 用帶標記的 package hash。Guarantor 把它對應到的 segments root 記入 B 的 report，accumulation 會把這張對照表的鍵列為依賴。若直接給 root，要指定先後就須另列 A 為 prerequisite。
>
> The tagged package hash does. The guarantor records it in B's segment-root lookup, whose keys are dependencies. With a direct root, B must list A as a prerequisite to require that order.

Q083 §21 §4 §13

Coretime 與 staking 的哪些部分屬於 Gray Paper 定義的狀態轉移，哪些政策留給上層系統？

What do the specified transitions support for coretime and staking, and which policies are left to higher-level systems?

> JAM 維護 core 授權、驗證者集合與活動紀錄。Coretime 如何出售、staking 如何依紀錄決定獎懲，屬於上層制度。
>
> JAM maintains core authorization, validator sets and activity records. Coretime sales and the staking policies that turn those records into incentives belong to higher-level systems.

Q084 §11 §6 §22 (background)

一份 report 只有一位 guarantor 簽名。同一個 core 的另一位 guarantor 正好是區塊作者。他簽的 Safrole block seal 能算第二份擔保嗎？

A report has one guarantor signature. Another guarantor assigned to that core authors the block. Can the author's Safrole seal count as the second guarantee signature?

> 不能。Block seal 簽的是省去 seal 欄位的 header。這位作者若要當第二位擔保人，必須另外簽署包含該 report hash 的擔保訊息。
>
> No. The seal signs the header with the seal field omitted. To supply the second signature, the author must separately sign the guarantee message for that report hash.

Q085 §12 §3

同一輪中，service 9 先執行完，送一筆 deferred transfer 給 B。Service 2 也送了兩筆給 B。下一輪 B 會依什麼順序收到這三筆？

In one round, service 9 finishes first and sends B one deferred transfer. Service 2 sends B two. In what order does B receive the three transfers next round?

> 先是 service 2 的兩筆，依原本送出的順序，最後是 service 9 的那筆。平行累積按發送端 service ID 由小到大串接各自的 transfer 序列。
>
> B receives service 2's two transfers in issue order, then service 9's. Parallel accumulation concatenates each sender's sequence in ascending service ID order.

同一套 PVM，三種執行情境

One PVM, different invocation contexts

Q086 §4 App. A

同一支程式在慢機器上跑了兩倍久。兩個節點為什麼仍應在相同位置耗盡 gas？

The same program takes twice as long on a slower computer. Why should both nodes exhaust gas at the same point?

> 給定相同的機器與主機環境、輸入及起始 gas，兩者依規定的操作成本扣費。計費取決於協定步驟與執行狀態，而非牆上時鐘。
>
> With identical machine and host contexts, input and initial gas, both apply the protocol's cost rules. Metering follows execution steps and state independently of wall-clock speed.

Q087 App. A

一個陌生服務嘗試寫入沒有寫入權限的記憶體。為什麼這也屬於共識規格？

Why must consensus specify what happens when unknown service code writes to an inaccessible page?

> PVM 定義自己的記憶體、頁面權限及相應的 fault 行為。獨立實作要觀察到同樣的機器結果，才可一致處理這次 invocation。
>
> The PVM defines its memory, page permissions and resulting fault behavior. Independent clients must observe the same machine outcome to handle the invocation consistently.

Q088 §9 App. A App. B

Refine 和 accumulate 跑在同一套 PVM 上。為什麼服務在兩個入口能對鏈上狀態做的事不同？

Why do refine and accumulate have different powers over chain state if both run on the PVM?

> 呼叫情境決定可用的 host calls 與資料視角。Refine 取得受限的歷史輸入，accumulate 的環境則提供鏈上狀態變更能力。
>
> Each invocation defines its host calls and data view. Refinement has restricted historical inputs, while accumulation receives capabilities to change on-chain state.

Q089 App. B

資料很大的時候，為什麼 invocation 讓服務按需取資料，而不在啟動時把所有輸入一次塞進 RAM？

Why let service code fetch large inputs on demand instead of loading everything into RAM at startup?

> 這樣可避免不受程式控制的大量初始配置與 gas 成本。服務透過對應 host call 讀取實際需要的資料範圍。
>
> It bounds initial allocation and avoids large gas costs before the program can act. The service can request the data ranges it actually needs through host calls.

Q090 §12 App. B

服務在 accumulate 中先做部分更新，再耗盡 gas。有 checkpoint 和沒有 checkpoint，會怎樣影響保留的效果？

How can a checkpoint affect what survives if an accumulation invocation later runs out of gas?

> 異常結束時會選用 exceptional context。Checkpoint 可把當時的效果存入該 context，之後尚未 checkpoint 的變更不會因此一併保留。
>
> An exceptional exit selects the exceptional context. A checkpoint can save effects there, while later changes that were not checkpointed do not automatically survive.

把狀態與資料變成可驗證的承諾

Encoding and Merkle commitments

Q091 App. C

兩個團隊對同一個 header 的欄位值完全同意，卻算出不同的 hash。除了 hash 函式，還要查哪裡？

Two teams agree on every header field but compute different hashes. Besides the hash function, what should they compare?

> 先比對送進雜湊函式的完整位元組。協定 codec 的欄位順序、整數表示與長度編碼都會影響承諾。
>
> Compare the exact bytes passed to the hash function. Field order, integer representation and length encoding in the protocol codec all affect the commitment.

Q092 App. C §11

一個 work-item 的 refine 成功但回傳空資料，另一個因 gas 耗盡而失敗。節點從 work-digest 的編碼怎麼分辨？

One work-item returns an empty blob from refine. Another runs out of gas. How can a node tell the outcomes apart in the encoded work-digests?

> 先看結果的標記。成功分支帶有長度和輸出，空資料的長度是零。Out-of-gas 有自己的錯誤標記，後面沒有輸出資料。
>
> Read the result tag. Success carries a length-prefixed blob, even when its length is zero. Out-of-gas has its own error tag and carries no blob.

Q093 App. D

兩台節點把服務存在不同資料庫、資料列插入順序也不同。它們如何仍能得到同一個 state root？

How can clients with different databases and insertion orders still obtain the same state root?

> 先依協定把狀態映成標準的 state keys 與值，再按同一棵 Merkle trie 的規則取根。相同邏輯狀態便有相同承諾。
>
> Both map the logical state to the specified keys and values, then apply the same Merkle trie construction. The commitment is determined by that mapping.

Q094 §7 App. D App. E

輕節點要查「某個 storage key 的值」和「某筆服務輸出有沒有出現在清單裡」，該分別對哪種根驗證？

Which commitment should a light client use for a storage value versus membership in a service-output list?

> 儲存值依 state key 對 state-trie root 驗證。清單成員則依位置與包含證明對輸出序列的 Merkle root 驗證。
>
> Verify a storage value at its state key against the state trie root. Verify list membership at its position against the output sequence's Merkle root.

Q095 §7 App. E

每個區塊都追加一個輸出根，為什麼用 Merkle mountain 結構適合維護這段歷史？

Why use a Merkle mountain structure for a history that gains an output root each block?

> 追加時可重用既有子樹，按需要合併 peaks，得到新的承諾。節點能持續延長歷史，也能替過去的項目提供證明。
>
> Appending reuses existing subtrees and merges peaks as needed to update the commitment. The history can grow while retaining support for proofs about earlier entries.

把各章接成一個可驗證的系統

Putting the protocol together

Q096 §6 §11 App. F

驗證者指派需要洗牌。如果每個節點各用自己的隨機產生器洗一次，其他人還怎麼驗證指派？

How can every node verify a shuffled assignment instead of drawing its own unrelated permutation?

> 協定指定相同的輸入順序、seed 和洗牌演算法。各自重算會得到同一個排列，因此同一份簽名能對上同一個指派。
>
> Nodes use the specified input order, seed and shuffle algorithm. They reconstruct the same permutation and can check a signature against the same assignment.

Q097 §6 §15 §18 App. G

簽署一份 guarantee、證明匿名出塊票券、簽署 BEEFY 承諾，為什麼需要三種不同的驗證方式？

Why verify a guarantee, an anonymous ticket and a BEEFY commitment in different ways?

> Guarantee 要辨認為 report 負責的簽署者。Ring VRF 證明票券來自合格集合並產生可驗證輸出，BEEFY 的 BLS 簽名則方便聚合供外部驗證。
>
> Guarantees identify signers responsible for a report. Ring VRFs prove eligible membership and a verifiable output for tickets. BLS signatures let external verifiers check aggregated BEEFY commitments.

Q098 §16 §11 App. H

驗證者已收到自己的 bundle shard，但少了一個 exported segment 的 shard。它能把這個 core 的 assurance bit 設為 1 嗎？

A validator has its bundle shard but is missing its shard for one exported segment. Can it set the assurance bit for that core?

> 不能，這個 bit 要留為 0。它必須持有自己的 bundle shard 和每個 exported segment 對應的 shard，並驗過它們的 Merkle proofs。
>
> No. That bit stays zero. The validator must hold its bundle shard and its shard for every exported segment, and verify their Merkle proofs.

Q099 §6 App. I

父區塊還在 epoch e，下一個區塊已是 e+2。即使 e 的票券已集滿，Safrole 會用那些票券還是 fallback keys 驗證 block seal？

The parent is in epoch e and the next block is in e+2. Even if the ticket accumulator in e was full, does Safrole verify the seal using those tickets or fallback keys?

> 用 fallback keys。只有剛好前進一個 epoch、父區塊已在票券截止後的時段，而且原票池已滿，才用那批票券。跳過整個 epoch 就不符合第一項。
>
> Use fallback keys. Using tickets requires a one-epoch advance, a parent in the closing period, and a full prior ticket accumulator. Skipping a whole epoch fails the first condition.

Q100 §17 §10 §19

一條尚未 finalized 的分支已累積某份 report，此時有人提出負面 judgment。何時要排除這條分支？投 GRANDPA 票前還須滿足什麼？

An unfinalized branch has accumulated a report that now gets a negative judgment. When is that branch disregarded, and what must hold before a GRANDPA vote?

> 一票負評會觸發全體稽核。節點得知至少三分之一驗證者判該 report 無效時，會排除已累積它的分支。GRANDPA 的投票目標還必須滿足本地 audited 與其他選鏈條件。
>
> One negative judgment triggers auditing by all validators. A node that sees at least one-third of validators judge a report invalid disregards branches that accumulated it. Its GRANDPA target must also satisfy the audit and chain-selection rules.